

A Compact Cayley Lemma and a Sidon Difference-Avoidance Theorem

Abstract

We prove a compactness lemma for dense Cayley graphs on cyclic groups of prime order whose non-trivial Fourier coefficients are non-positively biased. The lemma implies that such graphs contain cliques of every fixed size. As an application, we prove the following Sidon difference-avoidance statement: for every fixed integer M , all sufficiently large N have the property that every Sidon set $A \subseteq [N]$ admits a Sidon set $B \subseteq [N]$ of size M with

$$(A - A) \cap (B - B) = \{0\}.$$

1 Notation and preliminaries

All finite and compact abelian groups are written additively. Their dual groups are also written additively: the trivial character is denoted by 0, the inverse of a character γ is denoted by $-\gamma$, and $\gamma + \delta$ denotes pointwise multiplication of characters. If G is a finite abelian group and $a : G \rightarrow \mathbb{C}$, define

$$\hat{a}(\chi) = \mathbb{E}_{x \in G} a(x) \overline{\chi(x)} \quad (\chi \in \hat{G}).$$

Then

$$a(x) = \sum_{\chi \in \hat{G}} \hat{a}(\chi) \chi(x), \quad \sum_{\chi \in \hat{G}} |\hat{a}(\chi)|^2 = \mathbb{E}_{x \in G} |a(x)|^2.$$

For $G = \mathbb{Z}_p$ we identify $r \in \mathbb{Z}_p$ with the character $x \mapsto e^{2\pi i r x / p}$, so

$$\hat{a}(r) = \mathbb{E}_{x \in \mathbb{Z}_p} a(x) e^{-2\pi i r x / p}.$$

If G is a compact abelian group with Haar probability measure μ , and $f \in L^1(G)$, define

$$\hat{f}(\chi) = \int_G f(x) \overline{\chi(x)} d\mu(x) \quad (\chi \in \hat{G}).$$

We shall use the standard Pontryagin-duality facts that a compact abelian group is connected if and only if its discrete dual group is torsion-free, and that a character on a subgroup of a discrete abelian group extends to the whole group. The latter follows from the divisibility, equivalently injectivity, of $\mathbb{T}$ as an abelian group.

For a finite simple graph $H = (V, E)$ and a function $a : G \rightarrow \mathbb{C}$ on a finite abelian group, define the Cayley homomorphism density

$$t_H(a) = \mathbb{E}_{(x_v)_{v \in V} \in G^V} \prod_{\{v, w\} \in E} a(x_v - x_w).$$

For a compact abelian group G and $f \in L^\infty(G)$, define

$$t_H(f) = \int_{G^V} \prod_{\{v, w\} \in E} f(x_v - x_w) d\mu^{|V|}(x).$$

2 The compact Cayley lemma

Theorem 2.1 (Compact Cayley lemma). *Fix $\ell \geq 2$ and $\eta > 0$. There are constants $\varepsilon = \varepsilon(\ell, \eta) > 0$ and $p_0 = p_0(\ell, \eta)$ such that the following holds. Let $p > p_0$ be prime, and let*

$$T = -T \subseteq \mathbb{Z}_p, \quad 0 \notin T,$$

with $|T| \geq \eta p$. Suppose that

$$\widehat{\mathbf{1}_T}(r) \leq \varepsilon \quad (r \in \mathbb{Z}_p \setminus \{0\}).$$

Then the Cayley graph $\text{Cay}(\mathbb{Z}_p, T)$ contains a clique of size ℓ .

The proof is by contradiction. The next four lemmas provide the compact limiting argument.

2.1 Fourier extraction

Lemma 2.2 (Fourier extraction). *Let $p_n \rightarrow \infty$ be primes, put $G_n = \mathbb{Z}_{p_n}$ and $\Gamma_n = \widehat{G_n}$, and let $h_n : G_n \rightarrow [0, 1]$. After passing to a subsequence, there exist a countable torsion-free discrete abelian group Γ , its compact dual group $G = \widehat{\Gamma}$, and, for every $\gamma \in \Gamma$, a sequence of finite characters $\gamma_n \in \Gamma_n$, defined for all sufficiently large n , with the following properties.*

(i) *For every finite $E \subseteq \Gamma$, the map $\gamma \mapsto \gamma_n$ is injective on E for all sufficiently large n .*

(ii) *For every $\gamma \in \Gamma$, the limit*

$$a(\gamma) = \lim_{n \rightarrow \infty} \widehat{h_n}(\gamma_n)$$

exists.

(iii) *There is a unique $f \in L^2(G)$ such that*

$$\widehat{f}(\gamma) = a(\gamma) \quad (\gamma \in \Gamma).$$

(iv) *For each integer $q \geq 1$, there is a finite set $F_q \subseteq \Gamma$, with $0 \in F_q$, such that, for all sufficiently large n , every $\rho \in \Gamma_n$ satisfying*

$$|\widehat{h_n}(\rho)| > 1/q$$

is of the form $\rho = \gamma_n$ for some $\gamma \in F_q$. Moreover, if $\gamma \notin F_q$, then

$$|\widehat{f}(\gamma)| \leq 1/q.$$

(v) *The compact group G is connected.*

Proof. For $q \geq 1$, define the large spectrum

$$L_n(q) = \{\rho \in \Gamma_n : |\widehat{h_n}(\rho)| > 1/q\}.$$

By Parseval,

$$|L_n(q)|q^{-2} \leq \sum_{\rho \in \Gamma_n} |\widehat{h_n}(\rho)|^2 = \mathbb{E}_{G_n} |h_n|^2 \leq 1,$$

so $|L_n(q)| \leq q^2$.

Passing to a diagonal subsequence, we may label these large spectra compatibly. More precisely, there are finite sets

$$A_1 \subseteq A_2 \subseteq \cdots$$

and, for each $a \in A_* := \bigcup_q A_q$, a sequence of non-trivial characters $\lambda_{a,n} \in \Gamma_n$, such that for each fixed q and all sufficiently large n ,

$$L_n(q) \setminus \{0\} = \{\lambda_{a,n} : a \in A_q\}.$$

This is obtained by first passing to a subsequence on which the cardinalities of all $L_n(q) \setminus \{0\}$ are fixed, and then extending the labelling from level q to level $q+1$ inductively.

Let $\mathcal{F}$ be the free abelian group generated by A_* . Its elements are finite sums

$$w = \sum_{a \in A_*} m_a a.$$

For such a word, define the finite character

$$w_n = \sum_{a \in A_*} m_a \lambda_{a,n} \in \Gamma_n$$

for all sufficiently large n .

There are countably many words. Passing to one more diagonal subsequence, we may assume simultaneously that, for every word w , the assertion $w_n = 0$ has an eventual truth value, and the sequence $\widehat{h}_n(w_n)$ converges.

Let

$$R = \{w \in \mathcal{F} : w_n = 0 \text{ eventually}\}.$$

This is a subgroup of $\mathcal{F}$. Define

$$\Gamma = \mathcal{F}/R.$$

For $\gamma \in \Gamma$, choose a representing word w and set $\gamma_n = w_n$ for all sufficiently large n . This is well defined up to changing finitely many terms, because if two words represent the same element of Γ , their difference lies in R .

If $\gamma \neq \gamma'$ in Γ , then $\gamma - \gamma' \notin R$, so $\gamma_n \neq \gamma'_n$ eventually. Hence finite sets of distinct elements of Γ have eventually distinct finite lifts. This proves (i).

By construction, $a(\gamma) = \lim_n \widehat{h}_n(\gamma_n)$ exists for each $\gamma \in \Gamma$, proving (ii). If $E \subseteq \Gamma$ is finite, then, for all sufficiently large n , the lifts of elements of E are distinct. Therefore Parseval gives

$$\sum_{\gamma \in E} |a(\gamma)|^2 = \lim_{n \rightarrow \infty} \sum_{\gamma \in E} |\widehat{h}_n(\gamma_n)|^2 \leq 1.$$

Since E was arbitrary, $a \in \ell^2(\Gamma)$. Thus there is a unique $f \in L^2(\widehat{\Gamma})$ with Fourier coefficients $\widehat{f}(\gamma) = a(\gamma)$, proving (iii).

Let

$$F_q = \{0\} \cup \{\text{the class of } a \text{ in } \Gamma : a \in A_q\}.$$

The large-spectrum assertion follows from the construction. To prove the final assertion in (iv), suppose $\gamma \notin F_q$ and $|\widehat{f}(\gamma)| > 1/q$. Then, for all sufficiently large n ,

$$|\widehat{h}_n(\gamma_n)| > 1/q,$$

so $\gamma_n \in L_n(q)$. Hence $\gamma_n = \lambda_{a,n}$ for some $a \in A_q$ along an infinite set of n . Since there are only finitely many $a \in A_q$, and every equality $\gamma_n = \lambda_{a,n}$ has an eventual truth value after the diagonal extraction, we get $\gamma_n = \lambda_{a,n}$ eventually for some $a \in A_q$. Thus γ equals the class of a , contradicting $\gamma \notin F_q$.

It remains to show that Γ is torsion-free. If $m\gamma = 0$ in Γ , with $m \geq 1$, then $m\gamma_n = 0$ in Γ_n eventually. Since $\Gamma_n \cong \mathbb{Z}_{p_n}$ and $p_n \nmid m$ for all sufficiently large n , the group Γ_n has no non-trivial m -torsion. Hence $\gamma_n = 0$ eventually, so $\gamma = 0$ in Γ . Therefore Γ is torsion-free. By Pontryagin duality, $G = \widehat{\Gamma}$ is connected. This proves (v). $\square$

2.2 Equidistribution and boundedness of the limit

Lemma 2.3 (Equidistribution of finite lifts). *Let $E \subseteq \Gamma$ be finite, and let*

$$P(x) = \sum_{\gamma \in E} c_\gamma \gamma(x)$$

be a trigonometric polynomial on $G = \widehat{\Gamma}$. Define its finite lift by

$$P_n(x) = \sum_{\gamma \in E} c_\gamma \gamma_n(x) \quad (x \in G_n).$$

Then

$$\mathbb{E}_{x \in G_n} P_n(x) \longrightarrow \int_G P d\mu.$$

Moreover, if $P \geq 0$ pointwise on G , then $P_n \geq 0$ pointwise on G_n for all sufficiently large n .

Proof. Let $\Delta \leq \Gamma$ be the subgroup generated by E . Since Γ is torsion-free and Δ is finitely generated, $\Delta \cong \mathbb{Z}^d$ for some d . Choose a basis $\alpha_1, \dots, \alpha_d$ of Δ . Then $\widehat{\Delta} \cong \mathbb{T}^d$.

The restriction map $G = \widehat{\Gamma} \rightarrow \widehat{\Delta}$ is surjective, because every character on Δ extends to a character on Γ . Thus P factors as

$$P(x) = Q(\alpha_1(x), \dots, \alpha_d(x))$$

for a Laurent polynomial Q on $\mathbb{T}^d$.

Define

$$\Phi_n : G_n \rightarrow \mathbb{T}^d, \quad \Phi_n(x) = (\alpha_{1,n}(x), \dots, \alpha_{d,n}(x)).$$

For each non-zero $m = (m_1, \dots, m_d) \in \mathbb{Z}^d$, the character $m_1 \alpha_1 + \dots + m_d \alpha_d$ is non-trivial in Γ , so its finite lift is non-trivial for all sufficiently large n . Hence

$$\mathbb{E}_{x \in G_n} \Phi_n(x)^m = 0$$

for all sufficiently large n . By the Weyl criterion on $\mathbb{T}^d$, the measures $(\Phi_n)_* \mathbb{P}_{G_n}$ converge weakly to Haar measure on $\mathbb{T}^d$. Consequently,

$$\mathbb{E}_{G_n} P_n = \mathbb{E}_{G_n} Q(\Phi_n(x)) \longrightarrow \int_{\mathbb{T}^d} Q dm = \int_G P d\mu.$$

If $P \geq 0$ on G , then by surjectivity of $G \rightarrow \widehat{\Delta}$, the Laurent polynomial Q is non-negative on all of $\mathbb{T}^d$. Therefore $P_n(x) = Q(\Phi_n(x)) \geq 0$ for every $x \in G_n$ once the finite lifts are defined. $\square$

Lemma 2.4 (Basic properties of the limit). *Let $h_n : G_n \rightarrow [0, 1]$ and let f be the compact limit obtained in Lemma 2.2. Then*

$$0 \leq f \leq 1 \quad \text{a.e. on } G.$$

Moreover,

$$\int_G f d\mu = \lim_{n \rightarrow \infty} \mathbb{E}_{G_n} h_n$$

whenever the limit on the right exists. If each h_n is symmetric, then f is symmetric a.e. If, in addition,

$$\max_{\rho \in \Gamma_n \setminus \{0\}} \widehat{h}_n(\rho) \rightarrow 0,$$

then

$$\widehat{f}(\gamma) \leq 0 \quad (\gamma \in \Gamma \setminus \{0\}).$$

Proof. Let P be a trigonometric polynomial on G . By the Fourier conventions and the construction of f ,

$$\int_G f P d\mu = \lim_{n \rightarrow \infty} \mathbb{E}_{G_n} h_n P_n.$$

Indeed, for a character γ one has

$$\int_G f \gamma d\mu = \widehat{f}(-\gamma) = \lim_{n \rightarrow \infty} \widehat{h}_n(-\gamma_n) = \lim_{n \rightarrow \infty} \mathbb{E}_{G_n} h_n \gamma_n.$$

Linearity gives the claim for P .

If $P \geq 0$, Lemma 2.3 gives $P_n \geq 0$ eventually. Since $0 \leq h_n \leq 1$,

$$0 \leq \int_G f P d\mu \leq \int_G P d\mu$$

for every non-negative trigonometric polynomial P .

Now let $\varphi \geq 0$ be continuous. By Stone-Weierstrass, choose trigonometric polynomials Q_j converging uniformly to $\sqrt{\varphi}$. Then $|Q_j|^2$ are non-negative trigonometric polynomials converging uniformly to φ . Hence

$$0 \leq \int_G f \varphi d\mu \leq \int_G \varphi d\mu$$

for every non-negative continuous φ . This implies $0 \leq f \leq 1$ a.e.

Taking $P \equiv 1$ gives

$$\int_G f d\mu = \lim_{n \rightarrow \infty} \mathbb{E}_{G_n} h_n$$

whenever the latter limit exists.

Assume next that each h_n is symmetric. Then

$$\widehat{h}_n(-\rho) = \widehat{h}_n(\rho) \in \mathbb{R}$$

for every $\rho \in \Gamma_n$. Passing to the limit gives

$$\widehat{f}(-\gamma) = \widehat{f}(\gamma) \in \mathbb{R} \quad (\gamma \in \Gamma),$$

which is equivalent to $f(-x) = f(x)$ a.e.

Finally, if $\gamma \neq 0$, then $\gamma_n \neq 0$ for all sufficiently large n . Therefore

$$\widehat{f}(\gamma) = \lim_{n \rightarrow \infty} \widehat{h}_n(\gamma_n) \leq \limsup_{n \rightarrow \infty} \max_{\rho \neq 0} \widehat{h}_n(\rho) \leq 0.$$

□

2.3 Counting convergence

For $a : G \rightarrow \mathbb{C}$ on a finite abelian group, define the finite Cayley cut norm

$$\|K_a\|_{\square, G} = \sup_{|\phi|, |\psi| \leq 1} |\mathbb{E}_{x, y \in G} a(x - y) \phi(x) \psi(y)|.$$

Lemma 2.5 (Spectral control of the cut norm). *If $a : G \rightarrow \mathbb{C}$ on a finite abelian group, then*

$$\|K_a\|_{\square, G} \leq \sup_{\chi \in \widehat{G}} |\widehat{a}(\chi)|.$$

Proof. Let A be the convolution operator

$$A\psi(x) = \mathbb{E}_{y \in G} a(x - y)\psi(y).$$

The characters diagonalize A , and its $L^2 \rightarrow L^2$ operator norm is

$$\|A\|_{2 \rightarrow 2} = \sup_{\chi \in \widehat{G}} |\widehat{a}(\chi)|.$$

For $|\phi|, |\psi| \leq 1$,

$$|\mathbb{E}_{x,y} a(x - y)\phi(x)\psi(y)| = |\langle A\psi, \bar{\phi} \rangle| \leq \|A\|_{2 \rightarrow 2} \|\psi\|_2 \|\phi\|_2 \leq \|A\|_{2 \rightarrow 2}.$$

□

Lemma 2.6 (Counting convergence). *Let $h_n : G_n \rightarrow [0, 1]$ and $f : G \rightarrow [0, 1]$ be as above. For every finite simple graph H ,*

$$t_H(h_n) \longrightarrow t_H(f).$$

Proof. Let $H = (V, E)$, and put $e(H) = |E|$. Choose finite symmetric sets

$$B_1 \subseteq B_2 \subseteq \dots \subseteq \Gamma, \quad \bigcup_q B_q = \Gamma,$$

such that $F_q \subseteq B_q$ for every q , where F_q is the finite set from Lemma 2.2.

For each q , the subgroup $\langle B_q \rangle$ is finitely generated and torsion-free, hence isomorphic to $\mathbb{Z}^d$ for some d . Choose a finite non-empty set $Q_q \subseteq \langle B_q \rangle$ such that

$$\frac{|Q_q \cap (Q_q - \gamma)|}{|Q_q|} \geq 1 - \frac{1}{q} \quad (\gamma \in B_q).$$

For example, after identifying $\langle B_q \rangle$ with $\mathbb{Z}^d$, take a sufficiently large box.

Define the Fejer kernel on G by

$$K_q(x) = \frac{1}{|Q_q|} \left| \sum_{\gamma \in Q_q} \gamma(x) \right|^2.$$

Then $K_q \geq 0$, $\int_G K_q d\mu = 1$, and

$$\widehat{K}_q(\delta) = \frac{|Q_q \cap (Q_q - \delta)|}{|Q_q|} \quad (\delta \in \Gamma).$$

For all sufficiently large n , the finite lifts of elements of $Q_q - Q_q$ are distinct. Define

$$K_{n,q}(x) = \frac{1}{|Q_q|} \left| \sum_{\gamma \in Q_q} \gamma_n(x) \right|^2 \quad (x \in G_n).$$

Then $K_{n,q} \geq 0$, $\mathbb{E}_{G_n} K_{n,q} = 1$, and, for $\delta \in Q_q - Q_q$,

$$\widehat{K}_{n,q}(\delta_n) = \widehat{K}_q(\delta)$$

for all sufficiently large n .

Let

$$g_{n,q} = h_n * K_{n,q}$$

with normalized convolution on G_n , and let

$$g_q = f * K_q$$

with Haar convolution on G . Since $K_{n,q}$ and K_q are non-negative probability kernels, and $0 \leq h_n, f \leq 1$, we have

$$0 \leq g_{n,q} \leq 1, \quad 0 \leq g_q \leq 1.$$

We first compare h_n and $g_{n,q}$. For every $\rho \in \Gamma_n$,

$$\widehat{g}_{n,q}(\rho) = \widehat{h}_n(\rho) \widehat{K}_{n,q}(\rho).$$

If $|\widehat{h}_n(\rho)| \leq 1/q$, then

$$|\widehat{h}_n(\rho) - \widehat{g}_{n,q}(\rho)| \leq 1/q,$$

because $0 \leq \widehat{K}_{n,q}(\rho) \leq 1$. If $|\widehat{h}_n(\rho)| > 1/q$, then $\rho = \gamma_n$ for some $\gamma \in F_q \subseteq B_q$, and therefore

$$\widehat{K}_{n,q}(\rho) = \widehat{K}_q(\gamma) \geq 1 - 1/q$$

for all sufficiently large n . Hence again

$$|\widehat{h}_n(\rho) - \widehat{g}_{n,q}(\rho)| \leq 1/q.$$

By Lemma 2.5,

$$\|K_{h_n - g_{n,q}}\|_{\square, G_n} \leq 1/q. \quad (2.1)$$

A telescoping argument gives

$$|t_H(h_n) - t_H(g_{n,q})| \leq e(H) \|K_{h_n - g_{n,q}}\|_{\square, G_n} \leq \frac{e(H)}{q}. \quad (2.2)$$

Indeed, replace the edge factors one at a time. When the edge $\{u, v\}$ is replaced, all other variables are fixed; the remaining product is bounded by a function of x_u times a function of x_v , both bounded in modulus by 1, so the cut norm applies.

For fixed q , the functions $g_{n,q}$ are finite Fourier polynomials with frequencies in $Q_q - Q_q$:

$$g_{n,q}(x) = \sum_{\delta \in Q_q - Q_q} \widehat{h}_n(\delta_n) \widehat{K}_q(\delta) \delta_n(x)$$

for all sufficiently large n . The coefficients converge to those of

$$g_q(x) = \sum_{\delta \in Q_q - Q_q} \widehat{f}(\delta) \widehat{K}_q(\delta) \delta(x).$$

Expanding the product defining $t_H(g_{n,q})$, each summand is indexed by an assignment of a frequency $\delta_e \in Q_q - Q_q$ to each oriented edge e . The average over G_n^V is a product of character averages, one at each vertex. For each such finite assignment, the relevant vertex characters are finite lifts of fixed elements of Γ , and whether each is trivial has an eventual truth value. Hence finite character orthogonality gives

$$\lim_{n \rightarrow \infty} t_H(g_{n,q}) = t_H(g_q). \quad (2.3)$$

It remains to let $q \rightarrow \infty$. The kernels K_q form an approximate identity. Indeed, for every fixed $\gamma \in \Gamma$, one has $\widehat{K}_q(\gamma) \rightarrow 1$, since $\gamma \in B_q$ eventually. Therefore $P * K_q \rightarrow P$ uniformly

for every trigonometric polynomial P . Since trigonometric polynomials are dense in $L^1(G)$ and convolution by K_q is an L^1 -contraction,

$$\|f * K_q - f\|_{L^1(G)} \rightarrow 0.$$

Thus

$$\|g_q - f\|_{L^1(G)} \rightarrow 0.$$

Since $0 \leq g_q, f \leq 1$, another telescoping argument gives

$$|t_H(g_q) - t_H(f)| \leq e(H) \|g_q - f\|_{L^1(G)} \rightarrow 0. \quad (2.4)$$

Combining (2.2), (2.3), and (2.4), and then letting $q \rightarrow \infty$, proves $t_H(h_n) \rightarrow t_H(f)$. $\square$

2.4 Clique forcing on connected compact groups

Lemma 2.7 (Clique forcing). *Let G be a connected compact abelian group with countable dual group. Let $f : G \rightarrow [0, 1]$ be measurable and symmetric. Suppose*

$$\int_G f \, d\mu \geq \eta > 0$$

and

$$\widehat{f}(\chi) \leq 0 \quad (\chi \in \widehat{G} \setminus \{0\}).$$

Then, for every $\ell \geq 2$,

$$t_{K_\ell}(f) > 0.$$

Proof. Set $g = 1 - f$. Then $0 \leq g \leq 1$ a.e.,

$$\widehat{g}(0) = 1 - \int_G f \, d\mu \leq 1 - \eta,$$

and, for every non-trivial character χ ,

$$\widehat{g}(\chi) = -\widehat{f}(\chi) \geq 0.$$

Since f is symmetric and real-valued, all Fourier coefficients of f , and hence of g , are real. Thus every Fourier coefficient of g is non-negative.

We first show that the Fourier series of g is absolutely convergent. Let $E \subseteq \widehat{G}$ be finite and let $\delta > 0$. Since $\widehat{G}$ is discrete abelian, choose a finite non-empty Følner set $\Phi \subseteq \widehat{G}$ such that

$$\frac{|\Phi \cap (\Phi - \chi)|}{|\Phi|} \geq 1 - \delta \quad (\chi \in E).$$

This can be done inside the finitely generated subgroup spanned by E by taking large boxes.

Define

$$K_\Phi(x) = \frac{1}{|\Phi|} \left| \sum_{\psi \in \Phi} \psi(x) \right|^2.$$

Then $K_\Phi \geq 0$, $\int_G K_\Phi \, d\mu = 1$, and

$$\widehat{K_\Phi}(\chi) = \frac{|\Phi \cap (\Phi - \chi)|}{|\Phi|}.$$

Since $0 \leq g \leq 1$,

$$\int_G g K_\Phi \, d\mu \leq 1.$$

On the other hand, K_Φ is a finite trigonometric polynomial and all coefficients involved are non-negative, so

$$\int_G g K_\Phi d\mu = \sum_{\chi \in \widehat{G}} \widehat{g}(\chi) \widehat{K_\Phi}(\chi) \geq (1 - \delta) \sum_{\chi \in E} \widehat{g}(\chi).$$

Therefore

$$\sum_{\chi \in E} \widehat{g}(\chi) \leq \frac{1}{1 - \delta}.$$

Letting $\delta \downarrow 0$ and then taking the supremum over finite E gives

$$\sum_{\chi \in \widehat{G}} \widehat{g}(\chi) \leq 1. \quad (2.5)$$

Thus

$$g_c(x) = \sum_{\chi \in \widehat{G}} \widehat{g}(\chi) \chi(x)$$

converges absolutely and uniformly. It is continuous and positive definite, and it has the same Fourier coefficients as g , so $g_c = g$ a.e. Replace g by this continuous representative and replace f by $1 - g$. This changes f only on a null set, and hence does not change $t_{K_\ell}(f)$: for each pair $i < j$, the map $(x_1, \dots, x_\ell) \mapsto x_i - x_j$ pushes Haar measure on G^ℓ to Haar measure on G .

We now have g continuous, positive definite, and $0 \leq g \leq 1$ pointwise.

There are two cases.

Case 1: $g(0) < 1$. Then $f(0) = 1 - g(0) > 0$. By continuity, there is an open neighbourhood V of 0 such that $f > 0$ on V . Choose an open neighbourhood U of 0 with $U - U \subseteq V$. For every $(x_1, \dots, x_\ell) \in U^\ell$, all differences $x_i - x_j$ lie in V , and hence

$$\prod_{1 \leq i < j \leq \ell} f(x_i - x_j) > 0.$$

Since $\mu(U) > 0$, this gives $t_{K_\ell}(f) > 0$.

Case 2: $g(0) = 1$. Define

$$H = \{x \in G : g(x) = 1\}.$$

Since g is continuous, H is closed. We show that H is a subgroup. By (2.5), the vector

$$v_x = (\sqrt{\widehat{g}(\chi)} \chi(x))_{\chi \in \widehat{G}}$$

belongs to $\ell^2(\widehat{G})$ for every $x \in G$, and

$$g(x - y) = \langle v_x, v_y \rangle, \quad \|v_x\|_2^2 = g(0) = 1.$$

If $x \in H$, then

$$1 = g(x) = \langle v_x, v_0 \rangle.$$

Equality in Cauchy's inequality gives $v_x = v_0$. Hence, if $x, y \in H$, then

$$g(x - y) = \langle v_x, v_y \rangle = \langle v_0, v_0 \rangle = 1,$$

so $x - y \in H$. Thus H is a closed subgroup.

The subgroup H is proper. If $H = G$, then $g \equiv 1$, contradicting

$$\int_G g d\mu = 1 - \int_G f d\mu \leq 1 - \eta < 1.$$

For $1 \leq j \leq \ell - 1$, set

$$H_j = \{t \in G : jt \in H\}.$$

Each H_j is a closed subgroup. It is proper: since G is connected, $\widehat{G}$ is torsion-free, so multiplication by j on G is surjective. Indeed, the annihilator of jG is

$$\{\chi \in \widehat{G} : j\chi = 0\},$$

which is trivial; hence jG is dense and, being compact, closed. Therefore $jG = G$. If $H_j = G$, then $jG \subseteq H$, so $H = G$, contradiction.

A connected compact group cannot be a finite union of proper closed subgroups. Otherwise, by Baire's theorem, one of them would have non-empty interior; a subgroup with non-empty interior is open, and an open subgroup of a connected group is the whole group. Hence

$$G \neq \bigcup_{j=1}^{\ell-1} H_j.$$

Choose

$$t \notin \bigcup_{j=1}^{\ell-1} H_j.$$

Then $jt \notin H$ for $1 \leq j \leq \ell - 1$, so $g(jt) < 1$ and hence $f(jt) > 0$. Since f is symmetric, also $f(-jt) > 0$.

By continuity, choose an open neighbourhood U of 0 such that

$$f((i-j)t + u) > 0 \quad (1 \leq i < j \leq \ell, u \in U - U).$$

On the positive-measure set

$$(0 + U) \times (t + U) \times \cdots \times ((\ell - 1)t + U)$$

all factors $f(x_i - x_j)$ are positive. Therefore $t_{K_\ell}(f) > 0$.

Both cases give the desired conclusion. $\square$

2.5 Proof of the compact Cayley lemma

Proof of Theorem 2.1. Suppose the theorem fails for some fixed $\ell \geq 2$ and $\eta > 0$. Then for each n there are a prime $p_n \geq n$ and a set

$$T_n = -T_n \subseteq \mathbb{Z}_{p_n}, \quad 0 \notin T_n, \quad |T_n| \geq \eta p_n,$$

such that $\text{Cay}(\mathbb{Z}_{p_n}, T_n)$ is K_ℓ -free and, with $h_n = \mathbf{1}_{T_n}$,

$$\max_{r \in \mathbb{Z}_{p_n} \setminus \{0\}} \widehat{h}_n(r) \leq 1/n.$$

Apply Lemmas 2.2 and 2.4, passing to a subsequence if necessary. We obtain a connected compact abelian group G with countable dual and a measurable symmetric function $f : G \rightarrow [0, 1]$ such that

$$\int_G f \, d\mu \geq \eta, \quad \widehat{f}(\chi) \leq 0 \quad (\chi \neq 0).$$

By Lemma 2.7,

$$t_{K_\ell}(f) > 0. \tag{2.6}$$

On the other hand, Lemma 2.6 gives

$$t_{K_\ell}(h_n) \longrightarrow t_{K_\ell}(f).$$

For every n , $t_{K_\ell}(h_n) = 0$: tuples with repeated vertices have a zero factor because $0 \notin T_n$, and tuples with distinct vertices cannot form a clique because the Cayley graph is K_ℓ -free. Hence $t_{K_\ell}(f) = 0$, contradicting (2.6). $\square$

3 Application to Sidon difference avoidance

Definition 3.1. A finite set S of integers is called *Sidon* if all equations

$$a + b = c + d \quad (a, b, c, d \in S)$$

are trivial, in the sense that $\{a, b\} = \{c, d\}$. Equivalently, every non-zero difference $s - s'$ with $s, s' \in S$ occurs for at most one ordered pair (s, s') .

Lemma 3.2 (A greedy Sidon subset bound). *For every $M \geq 1$, there is an integer R_M such that every R_M -element set of integers contains a Sidon subset of size M . One may take*

$$R_M = 1 + (M - 1) + \binom{M - 1}{2} + 2(M - 1)\binom{M - 1}{2}.$$

Proof. We greedily build a Sidon subset. Suppose a Sidon set C of size $s < M$ has already been chosen from an ambient integer set X . We bound the number of values $x \in X$ that cannot be adjoined to C .

First, x cannot already lie in C , giving at most s forbidden values. Second, a new difference involving x cannot coincide with an old non-zero difference from $C - C$. Since

$$|C - C \setminus \{0\}| = 2\binom{s}{2},$$

this gives at most

$$s|C - C \setminus \{0\}| = 2s\binom{s}{2}$$

forbidden values. Third, two new differences may collide only if x is the midpoint of two distinct elements of C , giving at most $\binom{s}{2}$ further forbidden values.

For $s \leq M - 1$, the total number of forbidden values is at most

$$(M - 1) + \binom{M - 1}{2} + 2(M - 1)\binom{M - 1}{2}.$$

Thus, if $|X| \geq R_M$, there is always an admissible next choice until M elements have been chosen. $\square$

Theorem 3.3 (Sidon difference avoidance). *For every integer $M \geq 1$, there is $N_0 = N_0(M)$ such that for every $N \geq N_0$ and every Sidon set $A \subseteq [N]$, there exists a Sidon set $B \subseteq [N]$ of size M such that*

$$(A - A) \cap (B - B) = \{0\}.$$

Proof. Fix $M \geq 1$, and let $R = R_M$ be as in Lemma 3.2. Set

$$\ell = 8R.$$

Let $\varepsilon > 0$ and p_0 be the constants from Theorem 2.1 with clique size ℓ and density parameter $\eta = 1/2$.

Let N be sufficiently large in terms of M . By Bertrand's postulate, choose a prime

$$4N < p < 8N.$$

We also require $p > p_0$ and $CN^{-1/2} < \varepsilon$, where C is an absolute constant large enough for the estimate below. Embed $[N]$ in $\mathbb{Z}_p$ in the natural way.

Let $A \subseteq [N]$ be Sidon, and put $k = |A|$. Define

$$D_A = (A - A) \setminus \{0\}.$$

Since A is Sidon, its positive differences are all distinct. Hence

$$\binom{k}{2} \leq N - 1,$$

and therefore

$$|D_A| = k(k - 1) \leq 2N - 2. \quad (3.1)$$

Now define

$$T = \mathbb{Z}_p \setminus (D_A \cup \{0\}).$$

Because $p > 2N$, no non-zero difference of two elements of $[N]$ wraps around modulo p , so D_A is naturally and injectively viewed as a subset of $\mathbb{Z}_p$. The set T is symmetric and $0 \notin T$. By (3.1),

$$|T| = p - 1 - |D_A| \geq p - 1 - (2N - 2) = p - 2N + 1 > p/2.$$

We verify the Fourier condition. For $r \in \mathbb{Z}_p$, let

$$F_A(r) = \sum_{a \in A} e^{-2\pi i r a / p}.$$

For $r \neq 0$,

$$\sum_{x \in D_A} e^{-2\pi i r x / p} = |F_A(r)|^2 - k.$$

Since also $\sum_{x \in \mathbb{Z}_p} e^{-2\pi i r x / p} = 0$ and $0 \notin D_A$, we get

$$\sum_{x \in T} e^{-2\pi i r x / p} = -1 - \sum_{x \in D_A} e^{-2\pi i r x / p} = k - 1 - |F_A(r)|^2 \leq k - 1.$$

Thus, with normalized Fourier coefficients,

$$\widehat{\mathbf{1}_T}(r) \leq \frac{k - 1}{p}.$$

From $\binom{k}{2} \leq N - 1$ we have $k = O(\sqrt{N})$, while $p > 4N$. Hence

$$\frac{k - 1}{p} = O(N^{-1/2}) < \varepsilon$$

for N sufficiently large.

Theorem 2.1 applies, so $\text{Cay}(\mathbb{Z}_p, T)$ contains a clique

$$C \subseteq \mathbb{Z}_p, \quad |C| = 8R.$$

Equivalently,

$$(C - C) \setminus \{0\} \subseteq T. \quad (3.2)$$

Average over all cyclic intervals of length N in $\mathbb{Z}_p$. Each point of $\mathbb{Z}_p$ lies in exactly N such intervals, so some cyclic interval I of length N satisfies

$$|C \cap I| \geq |C| \frac{N}{p} > 8R \cdot \frac{1}{8} = R.$$

After translating modulo p , identify I with $[N]$, and let $X \subseteq [N]$ be the image of $C \cap I$. Then $|X| \geq R$.

If $x, y \in X$ are distinct, then the residue $x - y \pmod{p}$ lies in T by (3.2). Since $|x - y| < N < p/2$, this residue is the ordinary integer difference $x - y$, viewed modulo p . As T is disjoint from D_A , we obtain

$$(X - X) \cap (A - A) = \{0\}. \quad (3.3)$$

By Lemma 3.2, the set X contains a Sidon subset B of size M . Since $B \subseteq X$, (3.3) gives

$$(B - B) \cap (A - A) = \{0\}.$$

This proves the theorem. □